

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	99.39%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	94.22%	Share of actual attacks successfully flagged
F1-Score	96.74%	Balanced measure combining Precision and Recall
Accuracy	98.86%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.13%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	5.78%	Actual attacks that went undetected
True Positive Rate (TPR)	94.22%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.87%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 14907	FN 915
Actual: Normal	FP 91	TN 71998

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	99.99%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	100.00%	Share of actual attacks successfully flagged
F1-Score	99.99%	Balanced measure combining Precision and Recall
Accuracy	99.99%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	2.20%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	0.00%	Actual attacks that went undetected
True Positive Rate (TPR)	100.00%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	97.80%	Normal traffic correctly identified as safe

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	6987	0	0
Act: ICMP	0	2348	0
Act: UDP	0	0	5572

2c. Hybrid Cascaded

Metric	Value	Description
Precision	99.39%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	94.22%	Share of actual attacks successfully flagged
F1-Score	96.74%	Balanced measure combining Precision and Recall
Accuracy	98.86%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.13%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	5.78%	Actual attacks that went undetected
True Positive Rate (TPR)	94.22%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.87%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 14907	FN 915
Actual: Normal	FP 91	TN 71998

2d. Response Latency

Metric	Value	Description
Detection Time	342.6 ms	Average time taken to detect an attack
Mitigation Response Time	10.5 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2e. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	3.14%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	9.10%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	5.24%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.17	3	3	2026-09-25 12:14:49	2026-09-25 12:18:00
10.0.0.18	2	2	2026-09-25 12:18:05	2026-09-25 12:19:16
10.0.0.19	2	2	2026-09-25 12:18:06	2026-09-25 12:19:16
10.0.0.16	1	2	2026-09-25 12:18:10	2026-09-25 12:19:10
10.0.0.21	1	1	2026-09-25 12:18:36	2026-09-25 12:18:46
10.0.0.22	1	1	2026-09-25 12:18:37	2026-09-25 12:18:47
10.0.0.24	1	1	2026-09-25 12:19:07	2026-09-25 12:19:17
10.0.0.25	1	1	2026-09-25 12:19:06	2026-09-25 12:19:16

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-25 12:06:02	10.0.0.18	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:06:02	10.0.0.16	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:06:03	10.0.0.17	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:06:03	10.0.0.19	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:06:12	10.0.0.18	DDoS	SYN Flood	92.7%	Critical	Time Ban (1m)
2026-09-25 12:06:12	10.0.0.16	DDoS	SYN Flood	92.7%	Critical	Time Ban (1m)
2026-09-25 12:06:12	10.0.0.16	DDoS	SYN Flood	92.7%	Critical	Time Ban
2026-09-25 12:06:12	10.0.0.18	DDoS	SYN Flood	92.7%	Critical	Time Ban
2026-09-25 12:06:13	10.0.0.17	DDoS	SYN Flood	90.9%	Critical	Time Ban (1m)
2026-09-25 12:06:13	10.0.0.19	DDoS	SYN Flood	92.5%	Critical	Time Ban (1m)
2026-09-25 12:06:13	10.0.0.17	DDoS	SYN Flood	90.9%	Critical	Time Ban
2026-09-25 12:06:13	10.0.0.19	DDoS	SYN Flood	92.5%	Critical	Time Ban
2026-09-25 12:06:33	10.0.0.20	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:06:33	10.0.0.21	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:06:34	10.0.0.22	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:06:43	10.0.0.20	DDoS	UDP Flood	75.6%	High	Time Ban (1m)
2026-09-25 12:06:43	10.0.0.20	DDoS	UDP Flood	75.6%	Critical	Time Ban
2026-09-25 12:06:44	10.0.0.21	DDoS	UDP Flood	77.9%	High	Time Ban (1m)
2026-09-25 12:06:44	10.0.0.22	DDoS	UDP Flood	77.7%	High	Time Ban (1m)
2026-09-25 12:06:44	10.0.0.21	DDoS	UDP Flood	77.9%	High	Time Ban
2026-09-25 12:06:44	10.0.0.22	DDoS	UDP Flood	77.7%	High	Time Ban
2026-09-25 12:07:03	10.0.0.23	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:07:03	10.0.0.25	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:07:05	10.0.0.24	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:07:12	10.0.0.18	DDoS	SYN Flood	92.7%	High	Released
2026-09-25 12:07:12	10.0.0.16	DDoS	SYN Flood	92.7%	High	Released
2026-09-25 12:07:13	10.0.0.17	DDoS	SYN Flood	93.3%	High	Time Ban (2m)
2026-09-25 12:07:13	10.0.0.19	DDoS	SYN Flood	94.0%	Critical	Released
2026-09-25 12:07:13	10.0.0.23	DDoS	ICMP Flood	90.8%	High	Time Ban (1m)
2026-09-25 12:07:14	10.0.0.23	DDoS	ICMP Flood	90.8%	Critical	Time Ban
2026-09-25 12:07:14	10.0.0.25	DDoS	ICMP Flood	91.2%	High	Time Ban (1m)
2026-09-25 12:07:15	10.0.0.25	DDoS	ICMP Flood	91.2%	Critical	Time Ban
2026-09-25 12:07:15	10.0.0.24	DDoS	ICMP Flood	91.2%	Critical	Time Ban (1m)
2026-09-25 12:07:16	10.0.0.24	DDoS	ICMP Flood	91.2%	High	Time Ban
2026-09-25 12:07:23	10.0.0.19	DDoS	SYN Flood	94.0%	Critical	Time Ban (2m)
2026-09-25 12:09:58	10.0.0.18	DDoS	SYN Flood	93.4%	Critical	Time Ban (2m)
2026-09-25 12:11:58	10.0.0.19	DDoS	SYN Flood	92.7%	Critical	Time Ban (5m)

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-25 12:11:58	10.0.0.18	DDoS	SYN Flood	93.4%	Critical	Time Ban (5m)
2026-09-25 12:11:58	10.0.0.17	DDoS	SYN Flood	93.4%	Critical	Time Ban (5m)
2026-09-25 12:13:02	10.0.0.4	DDoS	Uncertain	59.6%	Low	Released
2026-09-25 12:13:02	10.0.0.9	DDoS	UDP Flood	92.0%	High	Time Ban (1m)
2026-09-25 12:13:08	10.0.0.10	DDoS	Uncertain	68.4%	Low	Released
2026-09-25 12:13:15	10.0.0.16	DDoS	SYN Flood	92.5%	Critical	Time Ban (2m)

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.24	ICMP Flood	0.6446	91.2%	High	Phase 2	1	10s	2026-09-25 12:19:17	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6288	93.8%	Critical	Phase 2	2	1m 11s	2026-09-25 12:19:16	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6271	94.0%	Critical	Phase 2	2	1m 10s	2026-09-25 12:19:16	Attack persisted - escalating to time ban
10.0.0.25	ICMP Flood	0.6456	91.2%	High	Phase 2	1	10s	2026-09-25 12:19:16	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6141	93.3%	High	Phase 2	2	1m 0s	2026-09-25 12:19:10	Escalated to Time Ban
10.0.0.22	UDP Flood	0.6402	77.7%	High	Phase 2	1	10s	2026-09-25 12:18:47	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6305	78.1%	High	Phase 2	1	10s	2026-09-25 12:18:46	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6460	93.7%	Critical	Phase 2	1	11s	2026-09-25 12:18:16	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6511	93.1%	Critical	Phase 2	1	10s	2026-09-25 12:18:16	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6380	93.1%	Critical	Phase 2	3	3m 11s	2026-09-25 12:18:00	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6380	93.1%	Critical	Phase 2	2	1m 11s	2026-09-25 12:16:00	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6380	93.1%	Critical	Phase 2	1	11s	2026-09-25 12:15:00	Attack persisted - escalating to time ban

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME
Network Administrator

DATE